

Investigating Small-Lesion Segmentation Failures in Acute Ischemic Stroke: An nnU-Net Baseline Analysis for ISLES’26

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Abstract. Automated segmentation of ischemic stroke lesions from MRI is a critical step toward scalable clinical decision support, yet performance on small and multifocal lesions remains a persistent bottleneck for existing deep-learning approaches. In this work, we present a baseline nnU-Net-based solution for the ISLES’26 challenge and conduct a size-stratified analysis of its failure modes. Using a 5-fold cross-validation protocol on the ISLES’26 training set (1453 cases), our baseline 3D full-resolution nnU-Net achieves an overall Dice score of 0.6465 (HD95 = 19.92 mm), but performance degrades substantially on small lesions (Dice = 0.3988) relative to large lesions (Dice = 0.7384). We characterize this bottleneck through a size-stratified evaluation and a dataset-level analysis showing that 66% of cases present with multiple, spatially distinct lesion components. We further report a preliminary architectural exploration using a residual-encoder U-Net variant, which improves large- and medium-lesion Dice but does not resolve the small-lesion deficit, suggesting that representational capacity alone is not the limiting factor. We conclude by outlining biologically motivated directions - including contralateral hemispheric symmetry features and vascular territory priors - as promising avenues for future work on this specific failure mode.

Keywords: Ischemic stroke, Lesion segmentation, nnU-Net, Deep learning, Small lesion segmentation, MICCAI ISLES 2026

1 Introduction

Acute Ischemic stroke is one of the major causes of long-term disabilities worldwide, and proper segmentation of infarcted areas with the help of MRI plays a fundamental role in making the diagnosis, planning the treatment, and predicting the outcomes [3]. Manual segmentation of the stroke lesions is a time-consuming procedure that varies between experts; therefore, the development of automated segmentation approaches remains a highly relevant task. Compared to previous versions of ISLES (especially ISLES’22-ATLAS containing 400 cases of stroke collected from multiple centers of which 250 were used for training), ISLES’26 brings about the major changes in terms of the amount of data and the diversity of the dataset itself: it is another expert annotated, multicenter MRI dataset for segmentation of stroke lesions in the similar way as it was for ISLES’22-ATLAS, but now on a larger scale. It contains approximately 2,000 scans of stroke taken from over 60 international centers, that makes ISLES’26 a dataset that is twice as large as ISLES’22-ATLAS in terms of the training data and the diversity of data increased by 40% compared to other ISLES challenges. In contrast to ISLES’22 where the focus was made only on the acute and sub-acute stroke cases imaged with the help of DWI, ADC and FLAIR sequences, ISLES’26 provides an open challenge for the segmentation of acute, sub-acute and chronic stages of the disease on T1 MRI. Every case is additionally annotated with the clinical data: number of days since stroke, binary variable indicating the chronicity of stroke and the information about the acquisition center of MRI scan. Thus, ISLES’26 is a dataset that is able to cover the gaps of ISLES’22-ATLAS challenge in terms of generalization ability of the models.

One such issue in the segmentation of stroke lesions is the relatively increased difficulty in detecting small and multifocal lesions. Small emboli and lacunar infarctions are quite frequent in real-world settings, are often seen to occur in conjunction with territorial infarctions in patients, yet are equally crucial for the proper evaluation of stroke severity. Conventional segmentation loss functions (like Dice, cross entropy) are

calculated using all the foreground voxels in the image/patch. This means that the model can obtain an excellent score while constantly failing to detect smaller lesions.

This paper seeks to:

1. Create an nnU-Net benchmark for the ISLES’26 segmentation problem via 5-fold cross-validation according to the nnU-Net v2 self-configuring methodology.
2. Examine the stratified performance of the benchmark model on the basis of lesion size, finding that small lesion segmentation is the key problem that is causing low accuracy.
3. Describe the lesion size distribution of the dataset.
4. Present an initial architecture design (residual encoder U-Net) and explain how higher model capacity does not address the problem with small lesions.
5. Present biological solutions to further pursue, including hemispheric symmetry properties and vascular territories.

2 Methods

2.1 Dataset

Our method uses the training set of the ISLES’26 dataset, which contains 1453 subjects and consists solely of T1-weighted images and associated binary segmentation masks of lesions. The detailed acquisition parameters (scanner field strength, TR/TE, native voxel size) of the ISLES’26 dataset, to our knowledge, have not been published until now, unlike ISLES’22-ATLAS, which provided a full table with the per-center acquisition settings of the dataset. Since the ISLES’26 dataset involves more than 60 international centers, which is more than an order of magnitude of center diversity compared to the 3 centers involved in the ISLES’22 dataset, considerable differences are to be expected in the dataset in terms of scanner vendor, field strength, and native resolution of the acquisition. Indeed, according to the organizers of the challenge, one of the goals of the challenge is to test the generalization of the model on real-world acquisition. Using nnU-Net dataset fingerprinting, we found out that the native resolution of the ISLES’26 dataset was around $1 \times 1 \times 1 \text{ mm}^3$ and used that resolution to standardize the acquisition of all volumes.

2.2 Data Characterization

In order to put this into context, we have examined the distribution of the lesion sizes and structures within the training data set. The lesion sizes range over several orders of magnitude, from tiny lacunar lesions to large territorial infarcts (Figure 1). By doing connected-component analysis on the ground-truth annotations, we found that about 66% of cases have two or more separate lesion components.

2.3 Baseline Segmentation Model

Our baseline segmentation model is built on nnU-Net v2 [1], a self-configuring framework that automatically derives network architecture, preprocessing, and training hyperparameters from dataset properties. We use the `3d_fullres` configuration with the framework’s default plain convolutional U-Net architecture (6 encoder stages; feature widths 32–64–128–256–320–320; instance normalization; leaky ReLU activation), trained with deep supervision across resolution scales. Training used a patch size of $128 \times 128 \times 128$ voxels, batch size of 2, foreground oversampling of 33% (nnU-Net default), and standard nnU-Net data augmentation (random flips, rotations, intensity perturbations). Each fold was trained for 500 epochs using stochastic gradient descent with a polynomial learning rate schedule (initial learning rate 0.01).

2.4 Cross-Validation Protocol

The baseline was assessed through 5-fold cross-validation, where the folds were automatically generated through the stratification process of nnU-Net. The predictions on the held-out fold validation for each fold were combined from the 5 folds, creating an estimation of the entire dataset’s performance (since each case is only predicted once).

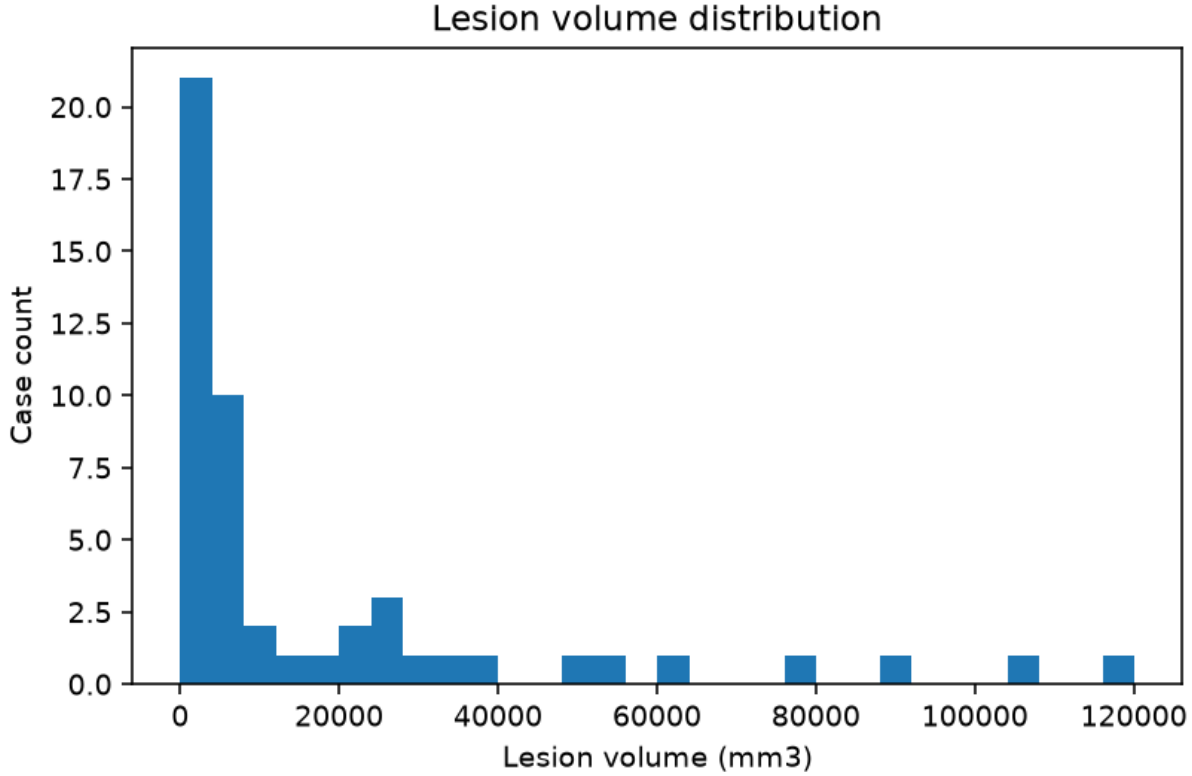


Fig. 1. Histogram of lesion volumes (mm^3) in a subset of the ISLES'26 training set. Clearly, the histogram is heavily skewed to the right, where most of the data points lie on the left side with lesion volumes under $10,000 \text{ mm}^3$, whereas on the other end of the spectrum, there are cases that have lesion volumes over $120,000 \text{ mm}^3$. This indicates that both small lesions, which could be lacunar infarcts, and large territorial infarcts co-exist in the dataset. Hence, the stratified analysis strategy is warranted in this study (Sect. 2), because otherwise, if we take one overall Dice score, it would either be biased toward small lesions or vice versa.

2.5 Evaluation Metrics and Size Stratification

We report the Dice similarity coefficient (DSC) and 95th-percentile Hausdorff distance (HD95) as primary metrics. To characterize size-dependent performance, we stratified cases into four groups based on ground-truth lesion voxel count: **EMPTY** (0 voxels, i.e., no lesion present), **SMALL** (10–498 voxels), **MEDIUM** (505–2993 voxels), and **LARGE** (≥ 3016 voxels). For cases where both the prediction and ground truth are empty (a true negative at the case level), Dice is defined as 1.0 rather than left undefined, following standard convention.

2.6 Preliminary Architectural Exploration

To determine if the small lesion deficit is due to the capacity constraint of the baseline architecture, we ran a preliminary experiment with nnU-Net’s residual-encoder U-Net preset (ResEnc-M) [2], where the plain encoder convolutional layers are replaced with residual blocks at equivalent computational cost, without any modification to the downsampling policy of the network. The model was trained randomly initialized with the `3d_fullres` setup mentioned above.

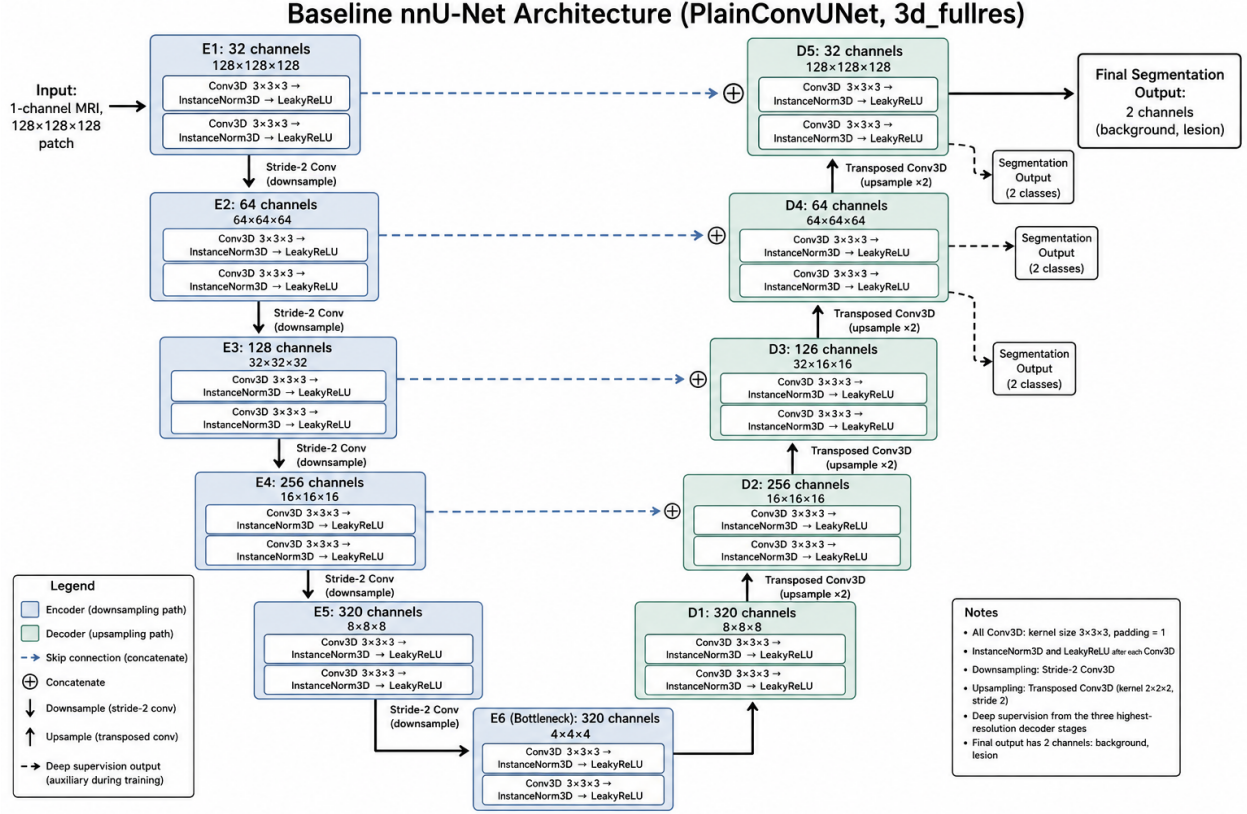


Fig. 2. Network Architecture for nnUnetv2 Baseline Model

3 Results

3.1 Baseline Cross-Validation Performance

Pooled 5-fold cross-validation results for the entire dataset ($n = 1453$) and lesion size stratification are presented in Table 1. The Dice score of the baseline on the whole set of images is equal to 0.6465 (HD95 = 19.92 mm). There is a considerable dependence of segmentation performance on lesion size. Large lesions have better segmentation results (Dice = 0.7384), whereas small lesions yield lower Dice scores (Dice = 0.3988).

From fold-wise analysis (Table 2), it can be observed that this effect is consistent across all folds and not a result of some specific split of the dataset, since Dice for SMALL lesions varies between 0.31 and 0.44, while Dice for LARGE lesions stays constant (0.71–0.75).

In addition, a certain number of cases of complete failure of segmentation (Dice=0), in spite of the presence of a non-negligible ground-truth lesion (size not SMALL); in the LARGE stratum there were 13 such cases. This means that a certain number of errors is due to total failure in localizing the lesions.

3.2 Preliminary Architectural Exploration (ResEnc-M)

Comparison between the residual-encoder version and the matched baseline is provided in Table 3. The residual-encoder architecture resulted in an increase in the Dice score for large (+0.011) and medium (+0.016) lesions while showing a slight decrease in the SMALL stratum (-0.016).

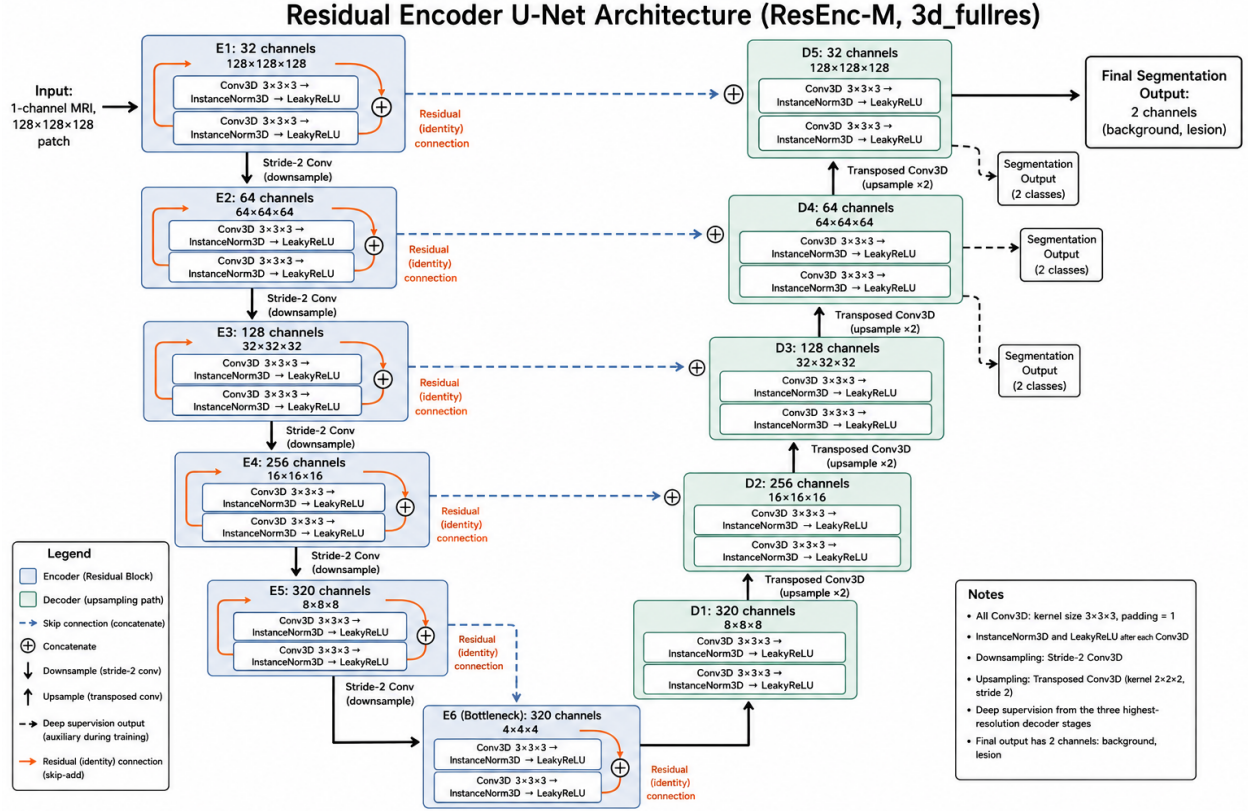


Fig. 3. Network Architecture for nnUnetv2 Basline Model + ResEncoder

4 Discussion & Conclusion

The findings suggest that small lesion segmentation is the main source of inefficiency in this baseline nnU-Net model for ISLES'26. The fact is in line with the properties of the data set: most instances (66%) have more than one lesions of different sizes, thus the standard loss functions, using voxel-wise pooling, do not encourage correct segmentation of small satellite lesions if there is a significantly larger lesion that dominates the loss.

Our architectural exploration indicates that this bottleneck is not due to the problem of network capacity, where an alternative residual encoder model with higher capacity per stage showed better performance for medium and large lesions, which require more representation at a fixed spatial resolution, but not for small lesions, whose performance was marginally reduced by using this model. This result seems to align with the downsampling scheme in our architecture, where small lesions occupy very little space compared to the volume at the bottleneck scale of the network.

We also observe that post-processing (taking the largest connected component in each case) did not help our experimentation due to the multifocal nature of the dataset, since the post-processing step would lead to the systematic elimination of true lesions instead of noise.

4.1 Future Work

Based on these findings, we identify two broad directions we believe are more promising than further architecture scaling for addressing the small-lesion bottleneck, motivated by the underlying pathophysiology of ischemic stroke rather than purely algorithmic changes:

- **Contralateral hemispheric symmetry features.** As ischemic lesions tend to be unilateral in nature, the generation of an asymmetry map based on comparison of each hemisphere to its opposite side might

Table 1. Baseline 5-fold cross-validation performance, pooled across all folds and stratified by ground-truth lesion size. n = number of cases in each stratum.

Stratum	n	Dice	HD95 (mm)
Overall	1453	0.6465 ± 0.2762	19.92
Large	882	0.7384 ± 0.2114	13.11
Medium	380	0.5604 ± 0.2759	27.10
Small	186	0.3988 ± 0.3210	38.09
Empty	5	0.2000 ± 0.4472	0.00

Table 2. Baseline per-fold Dice, stratified by lesion size.

Fold	Overall	Large	Medium	Small	Empty
0	0.6644	0.7511	0.5994	0.4277	0.0000
1	0.6599	0.7313	0.5696	0.4356	0.0000
2	0.6251	0.7067	0.5694	0.4401	1.0000
3	0.6468	0.7456	0.5829	0.3497	0.0000
4	0.6364	0.7548	0.4858	0.3129	0.0000

allow small, potentially hard to detect lesions to become much more apparent, by implicitly leveraging healthy tissue as a baseline reference.

- **Vascular territory and anatomical priors.** Ischemic lesions conform to the vascular territories of blood supply and, specifically in the case of small lacunar infarcts, concentrate in specific small vessel disease prone areas (such as the basal ganglia, internal capsule, or centrum semiovale). The use of a vascular territory atlas as an additional source of prior information (either as another component of the network or a post hoc anatomical filter for the identified small lesions) can offer a more focused approach for detecting small lesions compared to architecture-based methods.

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Table 3. Preliminary comparison of baseline (plain convolutional encoder) vs. residual-encoder (ResEnc-M) architectures.

Stratum	Baseline	ResEnc-M	Δ
Overall	0.6644	0.6734	+0.009
Large	0.7511	0.7621	+0.011
Medium	0.5994	0.6154	+0.016
Small	0.4277	0.4117	-0.016